Forecasting Monthly International Airline Passengers

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1 Introduction

In this project, the aim is to forecast the number of monthly international airline passengers using a time series dataset that covers a period from January 1949 to December 1960. The series records monthly passenger totals and provides a useful case study because it displays clear long term growth with a string repeating seasonal pattern. These characteristics make it suitable for comparing different forecasting approaches and assessing how well they capture the trend and seasonality.

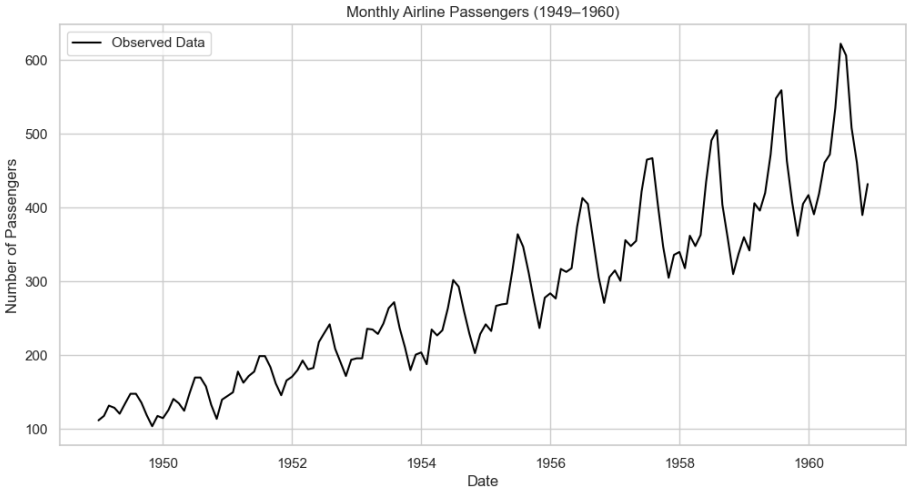


Figure 1: Monthly Airline Passengers (1949–1960)

The main objective of the project is to investigate how effectively different models can forecast future passenger numbers and to determine which approach performs best on this dataset. To compare, we considered three models. First, a seasonal naive model was used as a baseline, this provided a

benchmark by forecasting each month using the value observed in the same month of the previous year. The second was a multiplicative Holt-Winters model as a statistical forecasting approach, this was chosen because the data shows both trend and seasonality and the size of the seasonal fluctuations appears to increase as the series level rises. Finally an LSTM neural network was implemented as the deep learning model in order to test if a more flexible sequence-based method could improve forecasting accuracy.
The aim was to generate forecasts, but also compare the performance of the models from different categories.This is done to see if a simple seasonal naive is enough to provide a string forecast, or if the statistical model is a more suitable for the structure of our dataset, or if a deep learning approach can outperform the traditional methods on a small dataset. This is a useful comparison because it highlights the strengths and limitations of the different forecasting approaches rather than making an assumption that a complex model performs better.
To evaluate the models fairly, the data was split into training and test periods and their performances was measured using Root Mean Squared Error(RMSE). This allowed all our models to be assessed on the same out-of-sample observations. The goal was to identify the model with the lowest error and discuss why that model is the most appropriate for our dataset.

2 Analysis of Dataset

The dataset in the project is a univariate time series containing the number of international airline passengers from January 1949 to December 1960. This set has a total of 144 observations, recorded at regularly monthly intervals. Since data consists of a single variable observed over time, the main focus of the analysis is on the temporal structure of the series in particular its trend, seasonality and overall behaviour.
A visual inspection of the series shows that the data is clearly not stationary. The first obvious feature is the presence of a strong upward trend, as passenger numbers increase steadily over the sample period. This suggests that the average level of the series changes over time rather than remaining constant. In addition to this long term growth the series also shows a very clear and regular seasonal pattern, with repeated peaks troughs occurring each year. This is an important observation because it strongly influences the choice of forecasting model chosen for the analysis.
To examine the structure of the series more closely, a multiplicative decomposition was used. The decomposition confirms the conclusions drawn from the original plot. The trend component captures the long term increase in passenger numbers while the seasonal component highlights the regular yearly pattern present throughout the sample. The remainder component is comparatively smaller suggesting that a large proportion of the variation in the series can be explained by the combination of trend and seasonality. This supports the idea that structured forecasting approaches should perform well

on this dataset.

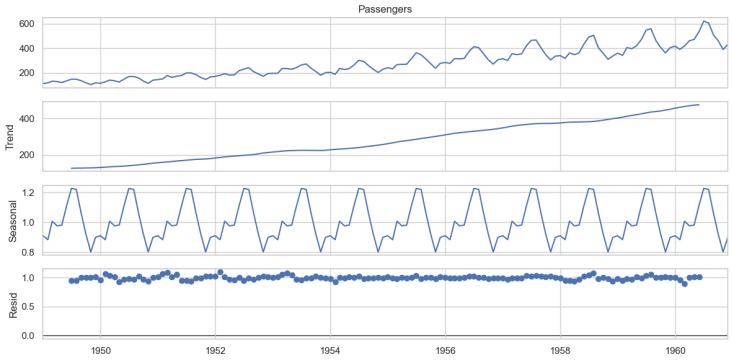


Figure 2: Multiplicative Decomposition

The dataset is also relatively small especially when compared with the amount of data typically used in deep learning applications. Although the series has strong and interpretable structure with limited number of observations means simple forecasting methods may have an advantage. This is one reason why its useful to compare a deep learning model with both a baseline method and a classical statistical method. In a dataset of this kind model complexity does not necessarily guarantee better forecasting performance.
Overall this dataset has several characteristics that make it suitable for forecasting analysis: its monthly, strong seasonal, trending, and consistent with a multiplicative seasonal structure. These features provide a clear justification for the methods selected in this project, namely seasonal naive benchmark, a multiplicative Holt-Winters and an LSTM model.

3 Methods

Three forecasting approaches were used in this project, a seasonal naive, a multiplicative Holt-Winters and an LSTM neural network model. These models were selected to represent the increase in complexity and to allow comparison between a simple baseline, a statistical method and a deep learning approach. Since the dataset is a monthly time series let y_t denote the number of passengers observed at time t, where the seasonal period $m = 12$.

3.1 Seasonal Naive Model

The first model used was the seasonal naive method, which served as the baseline forecast. For seasonal data, this method assumes that the value observed in a given month is similar to the value observed in the same month of the previous year. In other words, forecasts are obtained by repeating the last observed

seasonal cycle. Using the notation, the seasonal naive forecast can be written as

$$\hat{y}_{T+h|T} = y_{T+h-m(k+1)}$$

where m is the seasonal period and k is the integer part of $(h - 1)/m$. Since the airline-passenger data is monthly, $m = 12$. This means that for example the forecast for January next year is taken as the observed value from the previous January.

The seasonal naive model is useful because it provides a very simple benchmark against which more complex models can be judged. If an advanced model cannot improve on this baseline then its additional complexity is not justified. For a strongly seasonal dataset like this one, the seasonal naive is a very relevant benchmark.

3.2 Multiplicative Holt-Winters Model

The statistical model chosen for this project was the multiplicative Holt-Winters exponential smoothing model. This models appropriate when a time series contains both trend and seasonality and when the seasonal fluctuations increase proportionally with the level of the series. Since the airline-passenger dataset shows an upward trend together with increasing seasonal amplitude the multiplicative form is more suitable than an additive one.

The Holt-Winters model decomposes the series into three components: level, trend and seasonality. Let l_t denote the level component, $b - t$ the trend component, and s_t the seasonal component at time t. In the multiplicative case the updating equations can be written as

$$\begin{aligned} l_t &= \alpha \frac{y_t}{s_{t-m}} + (1 - \alpha)(l_{t-1} + b_{t-1}), \\ b_t &= \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}, \\ s_t &= \gamma \frac{y_t}{l_t} + (1 - \gamma)s_{t-m}. \end{aligned}$$

where α, β, and γ are the smoothing parameters satisfying values between 0 and 1. The corresponding forecast equation is

$$\hat{y}_{T+h|T} = (l_T + hb_T)\, s_{T+h-m(k+1)}.$$

The model was fitted to the training data, and the final 12 months were forecasted and compared with the observed test values. The multiplicative Holt-Winters specification was selected because it directly reflects the structure seen in the data: a non-constant level, a persistent trend and seasonal effects whose magnitude grows over time.

3.3 LSTM Model

To represent the deep learning approach, LSTM (Long Short Term Memory) model was implemented. LSTM are a type of recurrent neural network (RNN)

designed to model sequential data while overcoming the vanishing gradient difficulties that affect standard RNNs. They do this by maintaining a cell state that carries long-term information through the sequence together with gated mechanisms that control what information is forgotten, updated and passed forward.
In this project, the LSTM model was trained using sliding windows of the time series. The data was first divided into training, validation and test sets. Since neural networks are sensitive to scale the series was standardised using the mean and standard deviation of the training set. The model the used the previous 12 months as input in order to predict the following 12 months which is consistent with the yearly seasonal structure of the data.
The implemented network consisted of single LSTM layer with 32 units followed by a dense output layer. The LSTM receives an input sequence $(y_{t-11}, \ldots, y_t)$ and learns a nonlinear mapping to future values through its hidden states and gating structure, The model parameters were estimated by minimising the mean squared error loss function using the Adam optimiser. To reduce overfitting early stopping was used based on the validation loss. This allows the training process to stop automatically once the model no longer improves on unseen validation data.
Although the LSTM is more flexible than the statistical model, it is also more data demanding. For this reason it is not automatically expected to outperform simpler approaches on a small and highly structured series such as airline passengers. Its inclusion is useful since it allows direct comparison between deep learning and classical time series methods.

3.4 Forecast evaluation

To compare the three models fairly, forecasts were evaluated on out-of-sample observations using Root Mean Squared Error (RMSE) defined as

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}.$$

This measure was chosen because it is widely used in forecasting and gives greater weight to larger errors. A lower RMSE indicates better forecasting performance. Using the same metric for the three models ensure the comparison between methods is consistent and meaningful.

4 Results

4.1 Seasonal Naive Results

The seasonal naive model was first used as a baseline forecast. Since the airline-passenger series showed a strong yearly seasonal pattern, this benchmark is quite meaningful, because it simply repeats the values for the

same the months in the previous year. The forecast produced by this method followed the general seasonal structure of the test period reasonably well which confirms that seasonality is a dominant feature of the dataset.
The RMSE obtained by the seasonal naive model was 50.708. This value serves as the benchmark against which the more advanced models were compared. Although the seasonal naive method is extremely simple and requires no model fitting it still performs reasonably because of the regular seasonality in the data. However the size of the forecasting error suggests that relying only on previous seasonal cycle is not sufficient to fully capture the structure of the series.

4.2 Holt-Winters Results

The multiplicative Holt-Winters model was then fitted as the main statistical forecasting model. This model was chosen because the dataset contains both a clear trend and a strong seasonal pattern while the amplitude of the seasonal variation increases with the level of the series. These characteristics are consistent with a multiplicative trend-seasonal structure, making Holt-Winters a natural statistical choice.

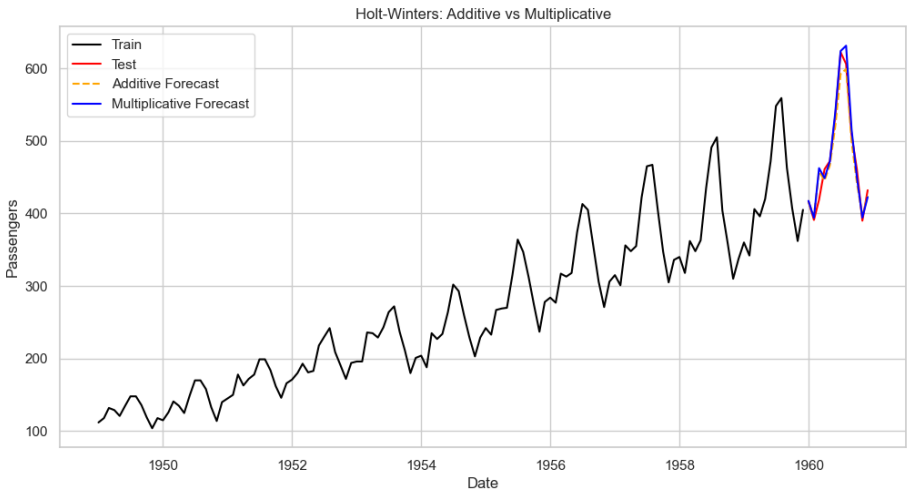


Figure 3: Holt-Winters Additive vs Multiplicative

The RMSE obtained by the Holt-Winters model was 15.810. This is substantially lower than the error of the seasonal naive model and indicated a major improvement in forecasting accuracy. The result shows that explicitly modelling the level, trend and multiplicative seasonal components leads to forecasts that are much closer to the observed test values.

4.3 LSTM Results

The LSTM model as implemented as the deep learning approach. Its purpose was to test whether a neural network model capable of learning nonlinear temporal relationships could outperform the more traditional forecasting methods. After training the model on the windowed and scaled data its forecasts were evaluated on the same test set as the other models.
The RMSE obtained by the LSTM model was 53.416. This was the highest RMSE among the three models meaning that the LSTM model produced the least accurate forecast. In this case the deep learning model did not improve on either the seasonal naive benchmark or the Holt-Winters model. This result is plausible because the airline passenger dataset is relatively small and has a strong, regular structure conditions under which classical time-series methods often perform better than more complex neural network models.

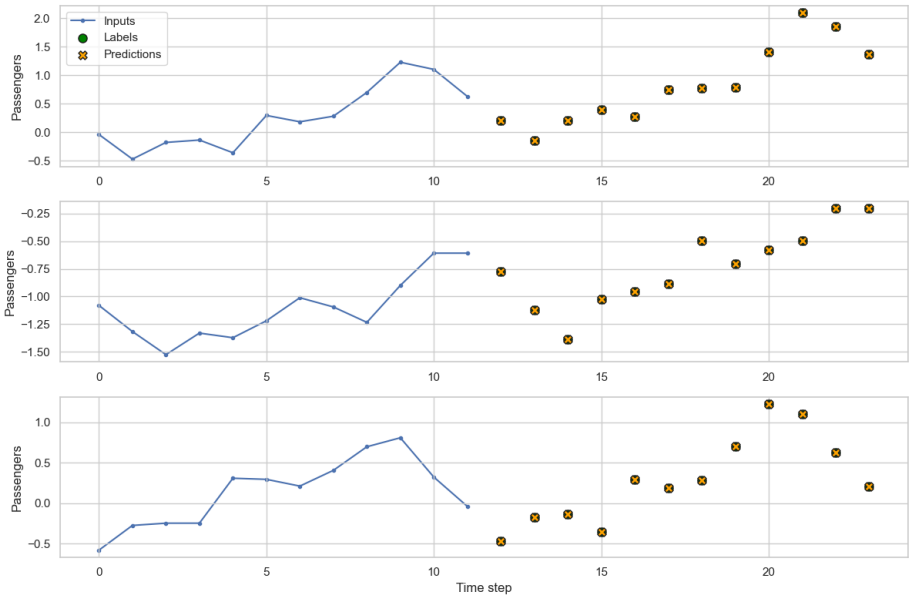


Figure 4: LSTM plot

4.4 Comparative discussion

The summarised comparison across the three models is as follows:-

- Seasonal naive RMSE: 50.708
- Holt-Winters RMSE: 15.810
- LSTM RMSE: 53.416

These results show clearly that the multiplicative Holt-Winters model was the most successful forecasting method for this dataset. Its RMSE was much lower than both the seasonal naive model and the LSTM model, indicating that it captured the underlying structure of the series far more effectively. This outcome is consistent with the earlier analysis of the data which showed a clear trend together with a strong multiplicative seasonality. Since Holt-Winters is specifically designed to model these features its superior performance is not surprising.
The comparison also shows that the seasonal naive baseline performed slightly better than the LSTM. This is important because it demonstrates that the added complexity of the deep learning model did not lead to better forecasts. This is shown as the LSTM did not outperform even the simple seasonal benchmark. This suggests that for a small dataset with a highly regular seasonal pattern a deep learning approach may be unnecessary and may even be less effective than much simpler alternative.

5 Conclusion

In this project the problem addressed was the forecasting of monthly international airline to passenger numbers using data from January 1949 to December 1960. The aim was to compare models from different categories in order to determine which method was most suitable for this dataset. A seasonal naive model was used as a baseline, a multiplicative Holt-Winters model was used as the statistical approach and an LSTM neural network was used as the deep learning model. All models were evaluated on the same test period using RMSE so that their forecasting performance could be compared fairly.
The results showed clear differences between the three approaches. The multiplicative Holt-Winters model performed best with an RMSE of 15.810 which was much lower than both seasonal naive of 50.708 and the LSTM at 53.416. This indicates that the Holt-Winters model was able to capture the main characteristics of the series much more effectively than the other two approaches. In particular, its strong performance reflects the fact the airline-passenger data has a clear upward trend and strong multiplicative seasonality, which are exactly the features that Holt-Winters is designed to model.
The comparison also provides useful insight into the strengths and limitations of the other methods. The seasonal naive model gave a reasonable benchmark because of the strong yearly seasonal pattern in the data, but it was too simple to account for the overall growth in the series. The LSTM model although more flexible produced the largest forecasting error. This suggests that for a relatively small and highly structured dataset, a deep learning model may not be the most effective choice. The result shows that increased model complexity does not automatically lead to better forecasts.

Overall the project shows that the most appropriate forecasting method depends on the nature of the data rather than on the complexity of the model. For this airline-passenger series, the multiplicative Holt-Winters model was the most suitable approach because it matched the observed trend and seasonal structure and produced the most accurate forecast. This makes it the preferred model for this forecasting problem.

6 Summary of AI use

ChatGPT was used to help with coding errors and indentation of coding to avoid issues when running the code. It was also used to help in writing report to fix grammatical issues and punctuation for a more smooth writing.